Zomato Restaurant Analysis – Complete Project Documentation

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Tools Used: Python, Pandas, NumPy, Power BI

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1. Project Title

Zomato Restaurants Analysis – Customer Trends, Service Impact & Strategic Recommendations

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2. Problem Statement

Bengaluru is one of India's most active food hubs with thousands of restaurants offering delivery, dining, and various service combinations.
Zomato generates large amounts of customer data including ratings, cost, services, locations, and restaurant types.

Business teams need answers to:

- What restaurant types are popular?
- How do services like delivery & table booking affect ratings?
- Which locations have the most restaurant density?
- What price ranges perform better?
- Which restaurants are the most reliable and most voted?
- Where is the business opportunity?

This project aims to **clean, analyse, and visualize** the Zomato dataset and generate **actionable insights**.

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3. Dataset Summary

Dataset: `zomato.xlsx` (40K+ restaurant records)

Key Columns:

- Restaurant name, location, rest type, cuisines
- Rate (0–5 scale)
- Approx_cost(for two people)

- Votes (customer engagement)
- Online_order / Book_table
- Dish_liked
- Listed_in(type), listed_in(city)

Derived Columns Added (Python):

- `rating_category`
- `cost_category`
- `ValueScore`
- `ReliableScore`
- Normalized service columns

Dataset cleaned using Python (`01_zomato_data_preprocessing.py`).

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4. Tools & Technologies

Python

- pandas
- numpy

Power BI

- DAX measures
- KPI Cards
- Bar Charts
- Column Charts
- Scatter plots
- Page navigation & filter controls

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5. Data Cleaning Summary (Python)

Key cleaning steps:

- Removed duplicates
- Cleaned rate field
- Converted cost & votes to numeric
- Normalized categorical fields
- Created rating & cost categories
- Handled missing values
- Generated ValueScore & ReliableScore

Result: **Clean dataset with 40K+ usable restaurant entries.**

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6. Dashboard Pages Summary

Dashboard has **6 pages**, each focusing on a core business dimension.

📄 Page 1 – Overview Dashboard

High-level metrics:

- **Total Restaurants:** 40K
- **Average Rating:** 3.69
- **Online Delivery Availability:** 65%
- Distribution of ratings
- Cost category distribution
- Location-wise restaurant count

Purpose:
Give management a quick understanding of Bengaluru's restaurant landscape.

📄 Page 2 – Location Analysis

Includes:

- Top 10 locations by density
- Top 10 by engagement (votes)
- Most expensive locations
- Location filters (dynamic)

Insights:
BTM, Koramangala, Indiranagar dominate both restaurant density & customer engagement.

📄 Page 3 – Ratings & Services Analysis

Examines customer experience:

- Rating distribution
- Online ordering impact
- Table booking impact
- Cost category vs rating

Insights:

- Online delivery improves rating (3.7 vs 3.6).
- Table booking gives highest rating (4.1 avg).
- Premium restaurants score 4.2+ but are very few.

📄 Page 4 – Restaurant Types & Cost Patterns

Focuses on format preferences:

- Most common restaurant types
- Best-rated restaurant types (count ≥ 50)
- Cost category distribution
- Cost vs rating bubble chart

Insights:
Customers love **Quick Bites, Cafe, Casual Dining**; premium formats are under-supplied.

📄 Page 5 – Best Restaurants & Top Chains

Highlights the top-performing brands:

- Best rated restaurant
- Best value for money
- Most popular (votes)
- Most reliable (Rating × Votes)
- Top 10 restaurant chains
- Top 10 restaurants by rating
- Top 10 by votes

Insights:
Onesta, CCD, Five Star Chicken dominate as chains.
Byg Brewski is the top-rated restaurant.

📄 Page 6 – Insights Summary & Recommendations

Your final presentation slide summarizing:

- Total restaurants
- Delivery availability
- Rating stabilization
- Low premium segment

- Location trends
- Service impact
- Cost behavior
- Strategic recommendations for:
    - Restaurant Owners
    - Investors
    - Data Teams

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7. Key Insights (Executive Summary)

⭐ Market Overview

- Bengaluru has **40K+ restaurants**, one of the largest markets in India.
- Customer ratings cluster between **3.5–4.2**.

⭐ Customer Behaviour

- Customers strongly prefer **affordable restaurants** (95% Low/Medium cost).
- **Premium restaurants are only 0.2%** but get **high ratings**.
- **Online ordering boosts ratings** (3.7 vs 3.6).
- **Table booking pushes ratings above 4.0**.

⭐ Location Insights

- BTM, Koramangala, Indiranagar are **top hotspots**.
- High engagement but high competition too.

⭐ Restaurant Types

- Quick Bites, Cafe, Casual Dining are the most common.
- Best-rated formats: **Microbrewery, Cafe, Iranian Cafe**.

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8. Business Recommendations

For Restaurant Owners

- Enable **online ordering** & **table booking**.
- Target high-density areas (BTM/Koramangala).
- Maintain average pricing **₹400–₹1500**.
- Improve cleanliness & ambience.

For Investors / Entrepreneurs

- **Premium dining is under-supplied** → huge opportunity.
- Best formats: Microbreweries, Cafe Lounges, Fine Dining.
- Franchise potential: **CCD, Onesta, Five Star Chicken**.

For Data & Analytics Teams

- Build Reliability Score = Rating × Votes.
- Track cuisine-wise gaps.
- Use clustering to identify new delivery zones.

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9. Conclusion

This Power BI project provides a **comprehensive, data-backed analysis** of Bengaluru's restaurant ecosystem.
The insights help restaurant owners, investors, and operations teams **make strategic business decisions** around:

- Pricing
- Restaurant type
- Locations
- Customer services
- Expansion planning

The final dashboard is a **powerful decision-support tool** with actionable intelligence.

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10. Project Files

- `01_zomato_data_preprocessing.py`
- `zomato_clean.csv` (clean dataset)
- Power BI dashboard file
- PDF exports